# Cheat sheet

$pH = -\log[H_3O^+]$

$pH = pK_a + \log\frac{[A^-]}{[HA]}$

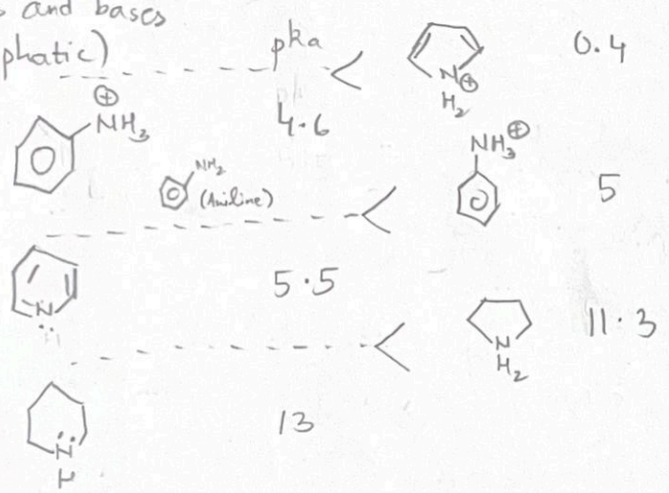
$K_a = \frac{[H_3O^+][A^-]}{[AH]}$

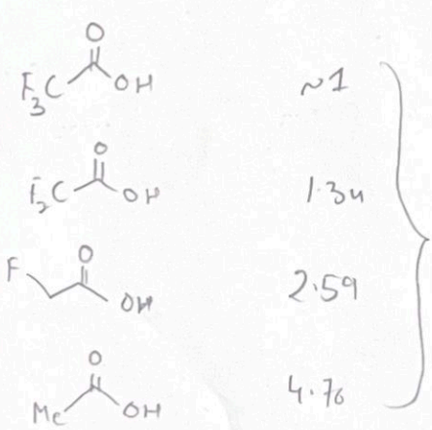
More delocalization of negative charge → Stable anion, lower pH (good LG)

| | | | |
|---|---|---|---|
| $HOCl$ | $ClO^-$ | 7.5 | |
| $HOClO$ | $ClO_2^-$ | 2 | |
| $HO-ClO_2$ | $ClO_3^-$ | −1 | |
| $HO-ClO_3$ | $ClO_4^-$ | −10 | |

Delocalization of lone pair (More acidic than basic)

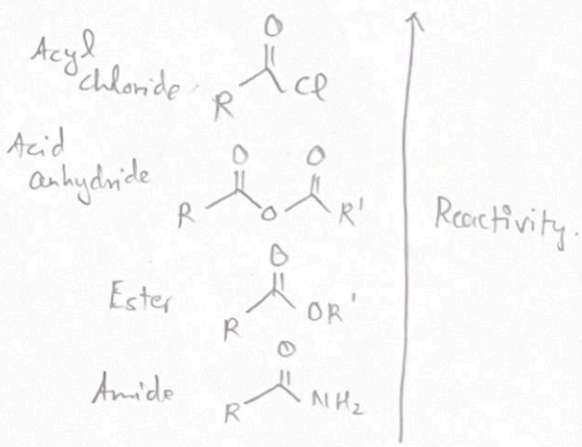
Nitrogen compounds act as acids and bases (aliphatic)

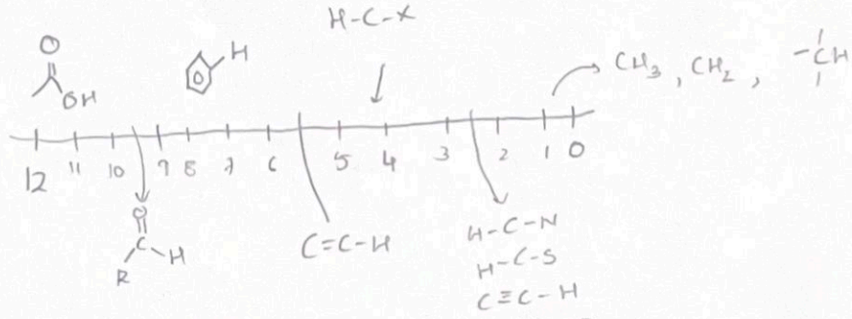


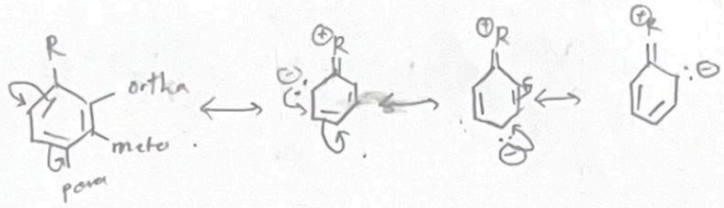




| Acid | pka | |
|---|---|---|
| $HI$ | −10 | |
| $HBr$ | −9 | |
| $HCl$ | −7 | |
| $H_2SO_4$ | −3 | |
| $H_3O^+$ | −1.7 | |
| | | 5 (enolate) |
| $H_2S$ | 7.0 | |
| $NH_4^+$ | 9.2 | (amide enolate) |
| | | 15 |
| $H_2O$ | 15.7 | |
| (Alcohols ~15) | | |
| $NH_3$ | 33 | |
| | | 19 (ketone enolate) |

| Basicity | Base | pka |
|---|---|---|
| $H_2CO_3$ | ($NaHCO_3$) | 5 |
| $H_2O$ | (KOH) | 15 |
| $H_2$ | (NaH) | 35 |
| Alkane | (BuLi) | 50 |



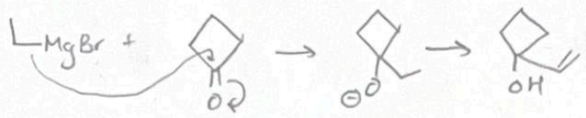



meta are more deshielded.

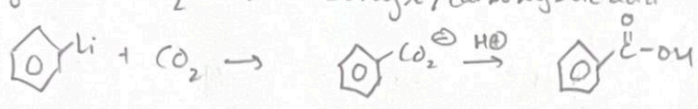
BO = $\frac{\text{Bonding} - \text{Antibonding}}{2}$

## Reactions

① Grignard + Carbonyl → Alcohol (Primary, sec, tertiary)




② Organo + $CO_2$ → Carbonyl / carboxyllic acid



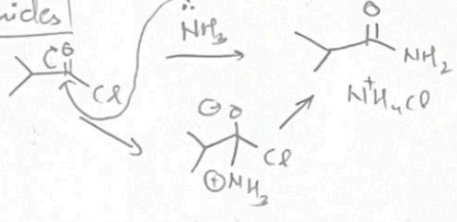

③ Oxidising agents

Primary Alcohol $\xrightarrow{[O]}$ Aldehyde $\xrightarrow{[O]}$ Carboxyllic Acid

Sec. Alcohol $\xrightarrow{[O]}$ Ketone

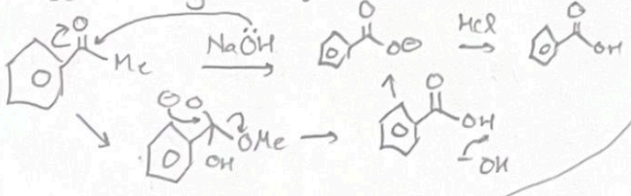
④ Carbonyls:

Forming
- **Amides**



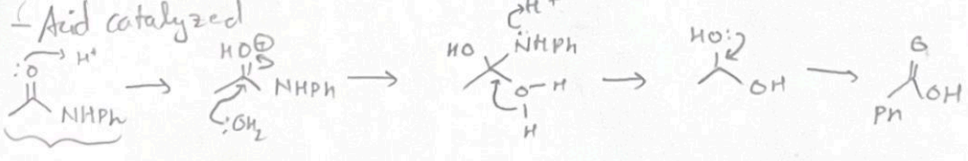

Forming
- **Carboxyllic acid**

+ Base - catalyzed hydrolysis



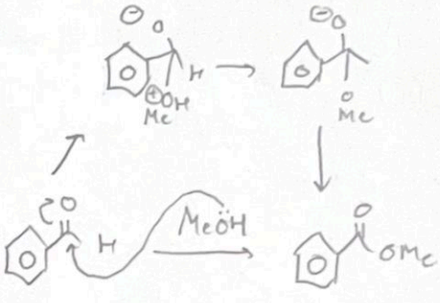

from + Amides

+ Acid catalyzed

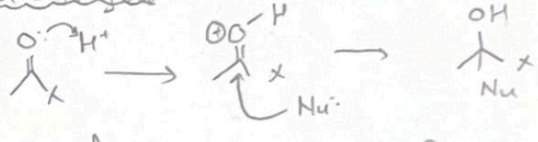



resonance stabilized

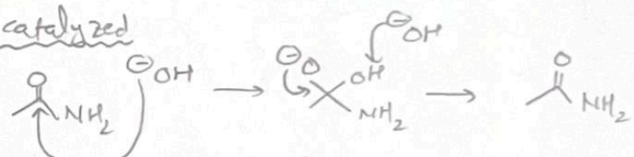
Forming
- **Esters**



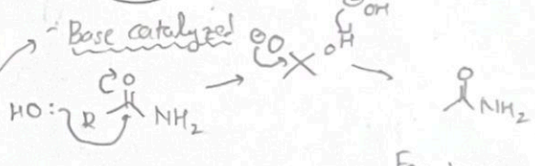

+ Acid catalyzed



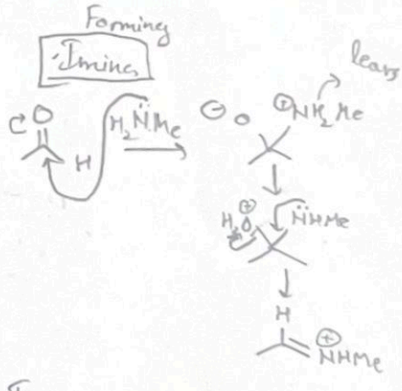

+ Base catalyzed



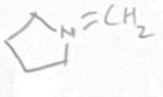

- Base catalyzed



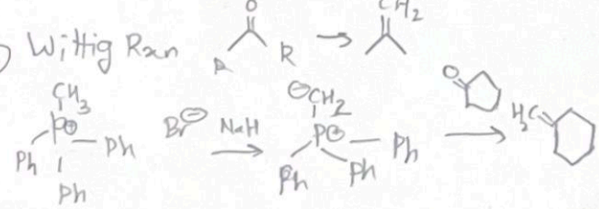

Forming
- **Imine**



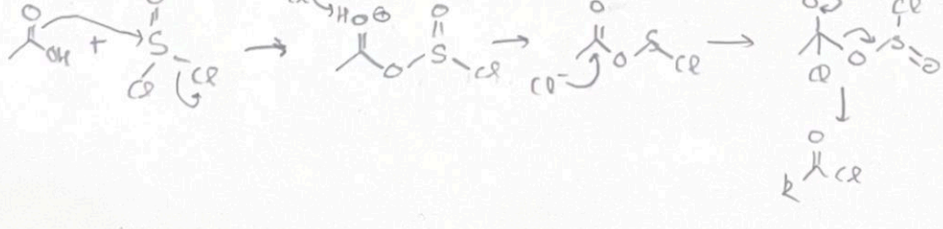

Iminium ion



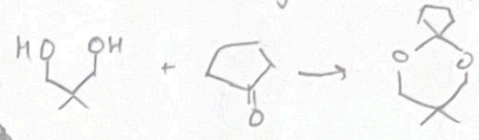

⑤ Wittig Rxn

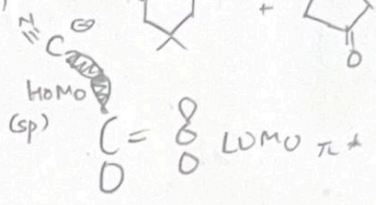



⑥



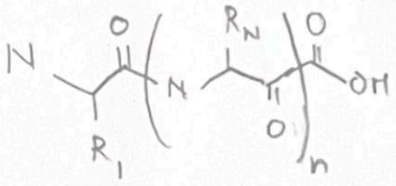

Alcohol + Carbonyl → Acetal



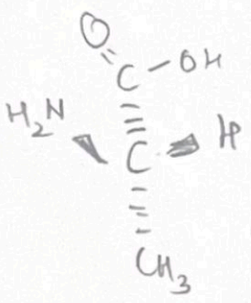


E 11

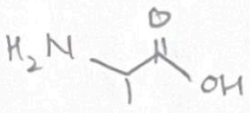
Proteins are large polymers made from amino acids.



General structure for proteins

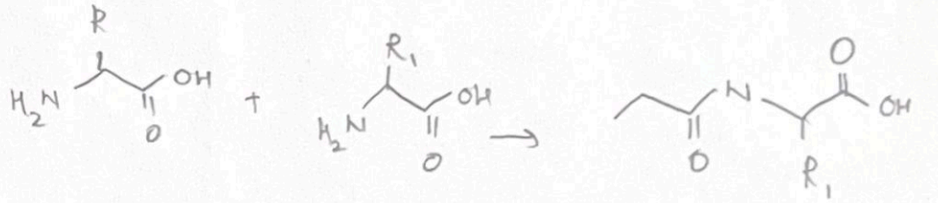


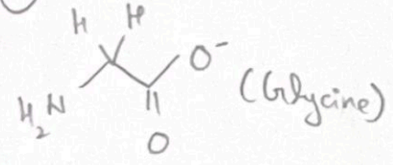
Alanine



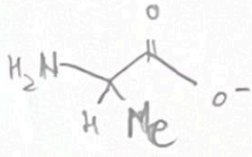
Amino Acid general structure.
↓
These are in L configura

Peptides & Proteins are made from amino acids via condensation.

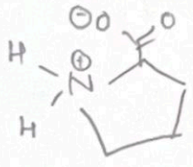


(1) 

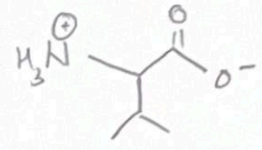
(Glycine)

(2) 

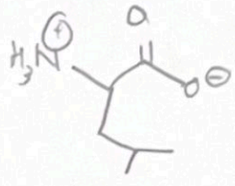
(Alanine)

(3) 

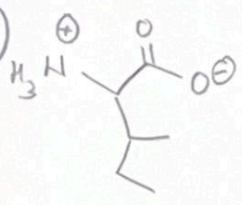
(Proline)

(4) 

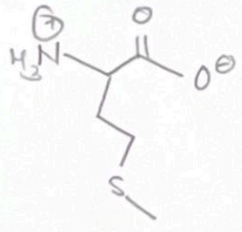
(Valine)

(5) 

(Leucine)

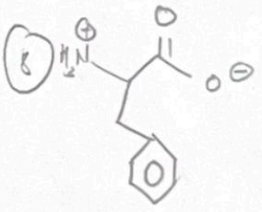
(6) 

(Isoleucine)

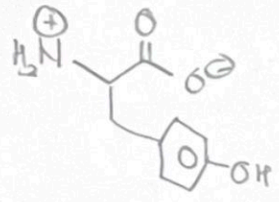
(7) 

(Methionine)

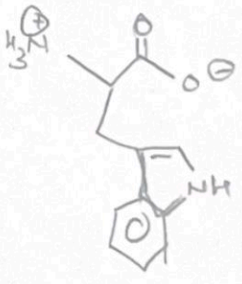
Hydrophobic

(8) 

(phenyl alanine)

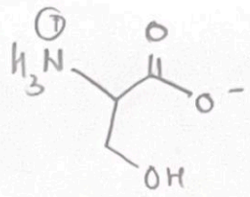
(9) 

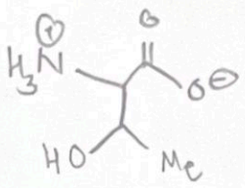
(Tyrosine)

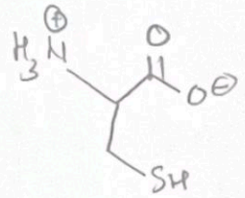
(10) 

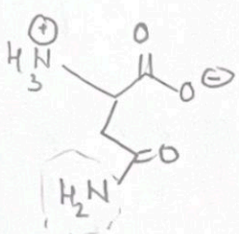
(Tryptophan)
↳ in Turkey.

Aromatic R groups

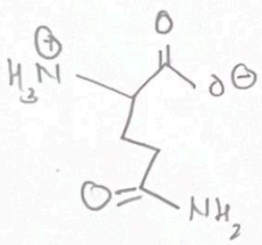
(11)  (Serine)

(12)  (Threonine)

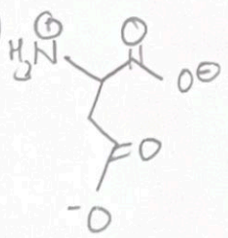
(13)  (Cysteine)

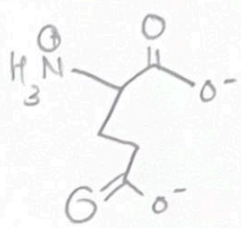
(14)  (Asparagine)

→ Neutral molecule.

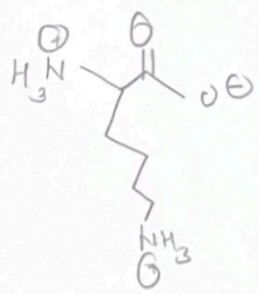
(15)  (Glutamine)

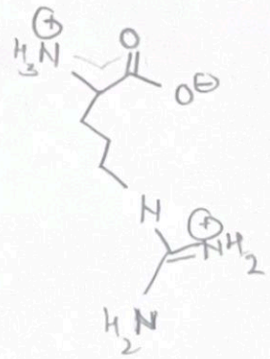
Polar, uncharged.

(16)  (Aspartate)

(17)  (Glutamate)

Negatively charged R groups

(18) 

(19) 

(20) 